

AI-Enabled Medical Device: LumineticsCore

AI/ML Component Description

LumineticsCore®, developed by Digital Diagnostics, is a software specifically designed for the automatic detection of diabetic retinopathy in retinal images. This software uses **deep learning algorithms** to analyze images captured from fundus cameras, determining the presence and severity of diabetic retinopathy without needing a specialist's input. The AI component likely relies on a dataset comprised of thousands of annotated retinal images, which may include both healthy and unhealthy conditions paired with clinical outcomes to enhance its accuracy.

Data for Training

The training data could include:

- Diverse retinal images from various demographics (age, sex, ethnicity) to ensure generalizability.
- Annotations from certified ophthalmologists regarding the presence, type, and severity of diabetic retinopathy.
- Clinical histories and other relevant patient data that aid in correlation (e.g., age, duration of diabetes).

Version 1.1, January 26th, 2026

Responsible AI License for LumineticsCore

Version 1.1, January 26, 2026

<http://licenses.lumi.ai/>

Section I: PREAMBLE

The **Responsible AI License** sets forth the terms governing the use of **LumineticsCore®**, a closed-source AI solution designed for automatic diabetic retinopathy detection. This License aims to ensure ethical usage by imposing restrictions that mitigate identified risks associated with the application of the Model, specifically in clinical settings.

Section II: DEFINITIONS

- **"Complementary Material"**: Source code and documentation required to run the Model, but not licensed under this agreement.
 - **"Contribution"**: Any updates or modifications made to the Model by authorized individuals.
 - **"Data"**: Any collected information, including retinal images and patient histories, used for training the Model.
 - **"Derivatives of the Model"**: Any modifications or variations derived from LumineticsCore.
 - **"Distribution"**: Sharing, reproducing, or hosting the Model in any form.
 - **"Harm"**: Physical, psychological, financial, or reputational damage inflicted on individuals or groups.
 - **"Licensor"**: The entity granting this License, retaining rights over the Model.
 - **"Model"**: The machine-learning assembly employed for analysis of retinal images.
 - **"Output"**: Results generated by using the Model.
 - **"User"**: Any person or entity utilizing LumineticsCore.
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Section III: USER QUALIFICATIONS

- Only **certified healthcare professionals** in ophthalmology may access LumineticsCore. Institutions must verify credentials before granting permission.
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Section IV: DATA USAGE AND PROTECTION

1. **Data Anonymization**: All patient data must be anonymized prior to use, ensuring compliance with GDPR and HIPAA regulations.
 2. **Informed Consent**: Users must obtain explicit consent from patients for any data utilized in training or quality improvement.
 3. **Quality Improvement**: Access to patient data is permitted solely for quality control and Model improvement, ensuring constant evaluation of AI performance.
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Section V: USE RESTRICTIONS

Users agree not to utilize the Model in the following manners:

1. **Legal Violations:** In any way that contravenes applicable laws or regulations.
 2. **Exploitation of Vulnerable Populations:** Targeting individuals, particularly minors or marginalized groups, in harmful or unethical ways.
 3. **Dissemination of Misinformation:** Generating or sharing content that could mislead patients or healthcare professionals.
 4. **Discriminatory Practices:** Engaging in practices that exacerbate healthcare inequities based on age, gender, ethnicity, or socioeconomic status.
 5. **Over-Reliance on AI Outputs:** Minimizing the role of clinical judgment in decision-making processes, especially in critical cases.
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Section VI: TRANSPARENCY OBLIGATIONS

1. **Disclosure:** Users must clearly inform patients about the AI's role in their diagnoses, including risks and limitations.
 2. **Documentation:** Maintain records of AI-generated outputs and any relevant treatments based on those outputs.
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Section VII: MONITORING AND ACCOUNTABILITY

1. **Error Reporting:** Users must document and report any misdiagnoses or discrepancies in performance.
 2. **Audits:** Regular audits will be conducted to ensure adherence to the License. Non-compliance may result in restrictions on model usage.
 3. **Continuous Improvement:** Users are required to participate in ongoing training related to the ethical use of AI technologies.
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Section VIII: LIMITATIONS ON MODIFICATIONS AND DISTRIBUTION

1. **Prohibition of Modifications:** Users may not modify the Model or its architecture without explicit written consent from the Licensor.

2. **No Redistribution:** The Model and its Derivatives cannot be distributed or shared without prior approval.

Section IX: INDEMNIFICATION

Users agree to indemnify and hold the Licensor harmless from any claims, damages, or legal actions arising from their use of LumineticsCore in violation of this License.

Section X: LIMITATION OF LIABILITY

The Licensor shall not be liable for any indirect, incidental, or consequential damages resulting from the use or inability to use the Model.

Section XI: TERMINATION

This License may be terminated by the Licensor upon a breach of any terms or conditions. Upon termination, all rights granted under this License shall cease.

Section XII: GOVERNING LAW

This License shall be governed by and construed in accordance with the laws of The State of Utah.
